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Improving attention-enhanced reservoir computing with Intrinsic Plasticity

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Dedicated to
my family and my friends.

Abstract

While the Transformer architecture has dominated the machine learning field this decade, there is a growing search for more efficient architectures during both training and inference. Reservoir computing, while remaining a strong class of models for time series prediction, struggles with discrete data like text. Attention-enhanced reservoir computing (AERC) tries to overcome this limitation using attention only on a fixed size reservoir, avoiding a growing KV cache. In this paper, we integrate Intrinsic Plasticity to the fixed reservoir to improve the AERC architecture and test it on a wider set of tasks. We find

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